



Assessing pS65 Ubiquitin levels in biofluids as a biomarker for Parkinson's Disease

Nicholas Hertz, Mitokinin Inc.

PPMI Project ID: 240

Summary

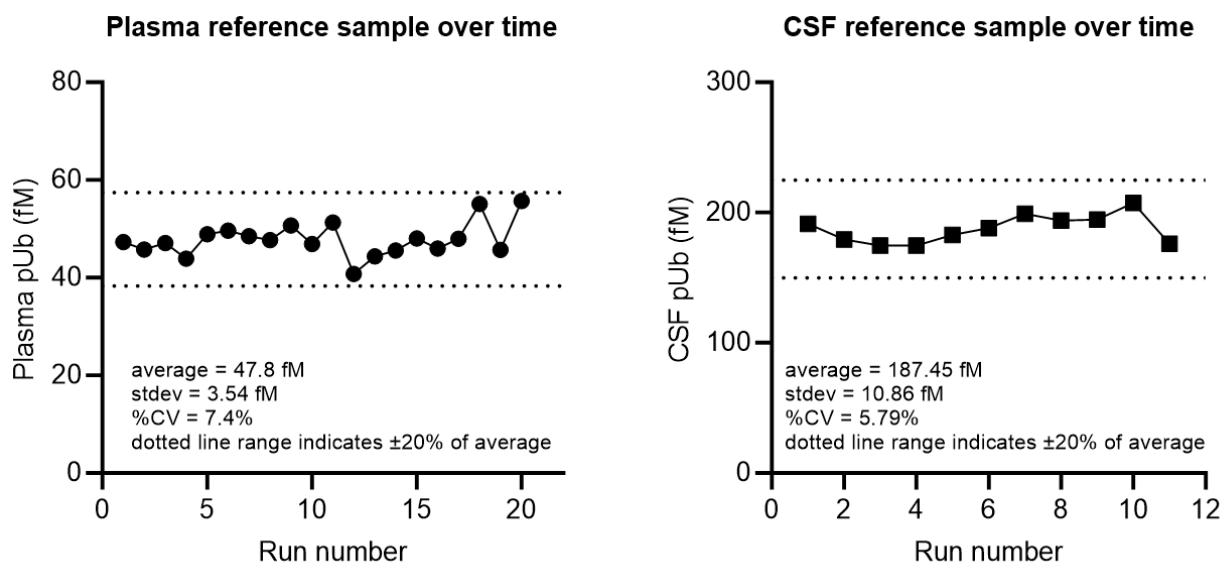
The PINK1/Parkin pathway is a key regulator of mitochondrial health, and loss of function mutations in PINK1/Parkin lead to early-onset Parkinson's Disease (PD). Upon mitochondrial damage, PINK1 is stabilized on the mitochondrial surface, phosphorylating serine 65 of Ubiquitin (pS65-Ub, or pUb) and leading to mitophagy downstream. PINK1 is the only kinase known to phosphorylate Ubiquitin. pUb can be elevated in cells with impaired mitophagy; this results in the extracellular release of pUb from cells¹. pUb is elevated in the brains and plasma of PD patients^{1,2}. Here, we quantify pUb levels in human plasma and CSF from healthy control and PD patients. Our results will (1) evaluate plasma pUb as a disease progression biomarker, (2) test if CSF pUb levels are significantly changed in the PD population, and (3) assess the correlation between plasma pUb and CSF pUb levels using matched samples.

Method

pUb levels were quantified using a pUb assay on the SMCxPRO platform using the manufacturer's recommended protocols and as described in ¹. As a capture antibody, rabbit anti-pS65-Ub antibody (CST clone E2J6T, catalog #62802, lot 5) was coated onto SMC magnetic beads using the Capture Antibody labeling kit (Millipore Sigma 03-0077-02). As a detection antibody, mouse anti-Ub antibody (CST clone P4D1, catalog #3936, lot 15) was labeled with Alexa Fluor dyes using the Detection Antibody labeling kit (Millipore Sigma 03-0076-02). For the calibration curve on every plate, K48 pUb tetramers (RnD systems UC-250) were used in a seven-point, two-fold dilution series from 344.9 fM to 5.4 fM. A blank sample was included on every plate as well. Plasma or CSF samples were added to the capture antibody beads on a 96 well, v-bottom plate, and mixed for 2 hours at 25°C on a microplate incubator/shaker (Jitterbug Model 130000, setting #4). Plate was placed on top of a Sphere Mag Plate (EMD 90-0003-02) and beads were washed using Millipore Wash Buffer (EMD 02-0111-03) on the Bio-Tek ELx™ 405 Microplate Washer (95-0004-05). Detection antibody was added to the beads and incubated for 1 hour at 25°C on microplate incubator/shaker (Jitterbug Model 130000, setting #5). Plate was again washed, and then elution and neutralization steps were performed using manufacturer's recommended protocol (Elution Buffer B (EMD 02-0297-00); Neutralization Buffer D (EMD 02-0368-00)). 17µL of neutralized eluate solution per well was transferred to the SMCxPRO™ reading plate (EMD Millipore PN02-1008-00; 384-well Aurora plate) and read on the SMCxPRO system (Millipore Sigma 95-0100-00). This assay has been validated for use in human plasma and CSF samples in parallelism, dilution linearity, and spike recovery experiments. Additionally, plasma and CSF from PINK1 KO animals have significantly lower pUb levels as compared to wildtype controls, confirming the specificity of the assay. The pUb assay was developed by Mitokinin, and will be commercially available at Millipore soon.

The figure below illustrates the stability of the pUb signal of the plasma and CSF reference samples supplied by PPMI. The inter-assay %CV was well below 10%, which indicates excellent assay stability and precision.





In total, excluding the reference samples shown in the above figure, 1321 plasma samples and 711 CSF samples were analyzed on this assay. For one of the CSF samples, the volume of CSF provided was insufficient for testing, and the TESTVALUE is denoted as "N/A."

References

1. Chin RM, Rakhit R, Ditsworth D, Wang C, Bartholomeus J, Liu S, Mody A, Laishu A, Eastes A, Tai C, Kim RY, Li J, Hansberry S, Khasnavis S, Rafalski V, Herendeen D, Garda V, Phung J, de Roulet D, Ordureau A, Harper JW, Johnstone S, Stöhr J, Hertz NT. Pharmacological PINK1 activation ameliorates Pathology in Parkinson's Disease models. bioRxiv [Preprint]. 2023 Feb 15:2023.02.14.528378. doi: 10.1101/2023.02.14.528378. PMID: 36824886; PMCID: PMC9949154.
2. Hou X, Fiesel FC, Truban D, Castanedes Casey M, Lin WL, Soto AI, Tacik P, Rousseau LG, Diehl NN, Heckman MG, Lorenzo-Betancor O, Ferrer I, Arbelo JM, Steele JC, Farrer MJ, Cornejo-Olivas M, Torres L, Mata IF, Graff-Radford NR, Wszolek ZK, Ross OA, Murray ME, Dickson DW, Springer W. Age- and disease-dependent increase of the mitophagy marker phospho-ubiquitin in normal aging and Lewy body disease. Autophagy. 2018;14(8):1404-1418. doi: 10.1080/15548627.2018.1461294. Epub 2018 Jul 28. PMID: 29947276; PMCID: PMC6372017.

About the Authors

This document was prepared by Jessica Li, Song Liu, Randall Chin, and Nicholas Hertz from Mitokinin, Inc. For more information, please contact Nicholas Hertz at nhertz@mitokinin.com.

Notice: This document is presented by the author(s) as a service to PPMI data users. However, users should be aware that no formal review process has vetted this document and that PPMI cannot guarantee the accuracy or utility of this document.

